

Health and plants: evidence checkpoint

Teacher guide and worked answers

Apply taught disease, immunity and plant concepts in an original diagnostic checkpoint.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsl/inputs/LAUNCH KS4 - 2026-27.xlsx | LAUNCH Weekly - Spring | C39

Health & plants assessment (Mar).

C39 consolidation of selected previously taught Topic5/6 content. Original16-mark classroom checkpoint after a short recap; not an official exam, grade predictor or a complete Topic5/6 assessment. Record the selected route and support.

Prior learning

Communicable disease, specific immune response and memory.

Photosynthesis, xylem/phloem and rate calculations.

Success evidence

Use causes and mechanisms rather than symptoms alone.

Explain a biological process with correct substances and structures.

Calculate a rate, identify a limitation and improve one answer.

Equipment

Shared evidence sheet, one selected response route, pen and ruler.

Projector or offline HTML/PowerPoint; calculator where useful.

Preparation

Print the shared page, one selected response route and the exit page. Routes are alternatives, not extra tasks.

Read the worked answers and note any constructed data before teaching.

Practical care

Paper and screen lesson; no human sampling, exercise test, breath holding, medicine handling or personal health disclosure.

Ready to teach alternative

All evidence is supplied. Use paper diagrams and cards, pointing, annotation or dictation if a device is unavailable.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	A checkpoint with a useful next step
2	Retrieval	2-4	Bring the core distinctions back
3	I do	4-7	Read causal evidence
4	I do	7-9	Keep the mechanism clear
5	I do	9-11	Calculate, then attach the unit
6	We do	11-13	Repair a worked explanation
7	We do	13-14	Check the role labels
8	We do	14-15	Audit a unit
9	Hinge	15-16	Ready to work independently?
10	You do	16-17	Choose the agreed route
11	You do	17-34	Independent checkpoint
12	Review	34-37	Feedback after the checkpoint
13	Review	37-39	Make the correction explain why
14	Exit	39-40	Plan the next five minutes of teaching
15	Exit	Within stage	Keep both versions

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening A checkpoint with a useful next step

Explain formative purpose, route recording and the 17-minute independent window. It is not a prediction of a GCSE grade. Keep pupil marks private.

2 Retrieval Bring the core distinctions back

Answers: no; no; amount/change and time with units. Do not reveal the checkpoint data calculations.

3 I do Read causal evidence

Model P communicable because a virus can spread; Q non-communicable because no infectious cause is transmitted. Use these analogous cases, not assessment A/B.

4 I do Keep the mechanism clear

Model one accurate sentence per role. Explain that the recap provides teaching support; checkpoint scores therefore describe post-recap performance.

5 I do Calculate, then attach the unit

Use 7, 8, 9 over 2 minutes: mean 8; rate 4 bubbles/min. Check that pupils see two different divisions with different purposes.

6 We do Repair a worked explanation

P's transmissible virus contrasts with Q's non-infectious cause. Accept a clear comparison without a diagnosis claim.

7 We do Check the role labels

Antibody. Ask how the memory-cell label should change; it is a cell. This is a warm-up, not the independent answer sheet.

8 We do Audit a unit

The 8 count was per 2 minutes; dividing by 2 gives 4 bubbles/min. Do not expose checkpoint 800-lux answer yet.

9 Hinge Ready to work independently?

Correct option: mean and rate answer different questions. Briefly re-teach timing if needed; record extra support.

A. A mean count and a per-minute rate can differ. - Yes. A count taken over several minutes must be divided by that duration.

B. Every mean count is automatically per minute. - Only if the count interval is one minute; check the supplied time.

10 You do Choose the agreed route

Read instructions only. A scribe/read-aloud may support access without adding science; record prompts or scientific help. Do not show answers during this window.

11 You do Independent checkpoint

Keep timer optional and discreet. Supported S1-S4, standard M1-M4 or challenge T1-T4 each totals 16 marks. No public competition. Collect a first response before feedback.

12 Review Feedback after the checkpoint

Now release relevant answers. Mark with the route-specific 16-mark scheme. Do not compare unlike support routes as equal test conditions.

13 Review Make the correction explain why

Allow private teacher/TA discussion. Retain first and revised response as evidence of learning; do not inflate the original independent score.

14 Exit Plan the next five minutes of teaching

Accept a specific accurate distinction and an actionable next step. Scores guide re-teaching by item, not a grade boundary.

15 Exit Keep both versions

Shared exit minute. Record route, support and one next teaching target.

Answers and assessment

Retrieval and warm-up

R1: no; R2: no; R3: amount/change per time with units. W1: P communicable/Q non-communicable because causes differ. W2: antibody protein; memory lymphocyte cell. W3: mean 8 is per 2 minutes; rate 4 bubbles/min. Hinge: mean and rate may differ.

S1 / M1 / T1 - 4 marks

1 mark: A communicable; 1 mark: B non-communicable; 1 mark: A has a transmissible viral cause; 1 mark: similar symptoms can have different causes. Do not credit an unsupported diagnosis.

S2 / M2 / T2 - 4 marks

1 mark: antigen is a recognised molecular feature triggering a specific response; 1 mark: antibody is a specific binding protein; 1 mark: memory lymphocyte is a cell; 1 mark: memory supports a faster later response to the same antigen. Accept equivalent accurate wording.

S3 / M3 / T3 - 4 marks

1 mark for both carbon dioxide and water; 1 mark: glucose; 1 mark: xylem for water/mineral ions; 1 mark: phloem for sucrose. Light is energy, not a material reactant; no credit for light replacing either reactant.

S4 / M4 / T4 - 4 marks

1 mark: $(30+33+36)/3=33$ bubbles; 1 mark: $33/3=11$ bubbles/min; 1 mark: valid control such as same shoot/solution/temperature/count duration; 1 mark: bubble sizes differ or oxygen dissolves. If an incorrect mean is carried forward, award rate-method mark for correctly dividing that mean by 3 with the correct unit.

L1 and exit

L1: credit a specific corrected explanation with reason, recorded separately from first score. E1: any accurate taught distinction. E2: actionable item-level target. A score alone is not sufficient diagnostic feedback.

Assessment record

Total 16 marks per route. Record route, read-aloud/scribe support and any scientific prompting. Do not assign official grades or claim qualification evidence completion. Re-teach: Q1 cause/route; Q2 cell/protein/memory; Q3 reactants/tissues; Q4 mean versus rate/proxy.

Teaching and assessment notes

40-minute lesson. Zero-minute slides share the preceding stage allocation.

Choose one response route. Accept accurate speech, pointing with explanation, annotations or writing; record the support used.

A teacher or TA can be a quiet audience: ask one evidence question, then let the pupil revise or justify a decision.

Suggested timing: opening 2, retrieval 2, I do7, We do4, hinge 1, independent 18 including instructions, feedback 5, exit 1. The independent response window is 17 minutes.

All routes use the same core items and 16-mark scheme; supported wording/word bank changes access. Record that difference and any additional prompting rather than treating scores as standardised comparisons.

Interactive warm-up is restricted to analogous rash cases, role labels and 7/8/9 counts over 2 minutes. It contains none of the assessment's constructed 800-lux calculation answers.

Access and responsive teaching

Supported

Same core concepts with chunked prompts and read-aloud access. Record support used; a pathway is not a diagnosis or fixed attainment label.

Standard

Explain the supplied evidence independently using accurate scientific terms and units.

Stretch

Optional deeper transfer or evaluation. Higher-tier-only material is labelled explicitly and is not required for the core outcome.

Regulation

Offer solo or partner work, a quiet response, a pass and brief reset with clear re-entry. No forced disclosure or public performance.

Misconceptions to check

A symptom proves the disease cause.

Similar symptoms can result from different causes.

Check: Use the two supplied cases without guessing a real diagnosis.

Antibodies are cells.

Antibodies are proteins; memory lymphocytes are cells.

Check: Ask pupils to distinguish the two objects.

Light becomes plant matter.

Light supplies energy; carbon dioxide and water are the material reactants.

Check: Check the reaction labels.

A bubble count is an exact oxygen volume.

Bubble size and dissolved gas limit the proxy.

Check: Ask what was actually counted.

Word	Meaning
pathogen	A disease-causing agent.
antigen	A feature recognised by the immune system that can trigger a specific response.
antibody	A protein that binds to a specific antigen.
memory lymphocyte	A cell supporting a faster later response to a previously encountered antigen.
proxy	A measurement used to indicate something else, with limitations.
diagnostic checkpoint	A task used to identify specific next teaching needs.

Scientific references

[Pearson Edexcel GCSE Biology 1BI0 specification](#)

Used to check relevant Foundation curriculum content; this is teacher-authored practice, not an official assessment or complete unit. Checked 9 September 2026.

WHO: How do vaccines work?

Checked specific recognition and immune memory; all checkpoint wording is original. Checked 9 September 2026.

SAPS: Measuring the rate of photosynthesis

Checked bubble-count interpretation. All assessment counts are constructed. Checked 9 September 2026.